

Registration Congruence Audit

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1 Scope

This report compares manuscript, protocol, and ClinicalTrials.gov registration signals when a ClinicalTrials.gov NCT identifier is available. Registry-derived findings are screening signals: they identify discrepancies or missingness that require source-document review, not adjudicated protocol violations.

2 Load Outputs

```
repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
study_id <- Sys.getenv("FORENSICS_STUDY_ID", "lungtime")
report_dir <- file.path(repo_root, "reports", "registration", study_id)
data_dir <- file.path(repo_root, "data", "processed", "registration", study_id)
```

```

required <- c("registration_row_results.csv", "registration_summary.csv")
missing <- required[!file.exists(file.path(report_dir, required))]
if (length(missing) > 0) {
  stop("Missing files: ", paste(missing, collapse = ", "))
}

row_results <- read.csv(file.path(report_dir, "registration_row_results.csv"), check.names = FALSE)
summary_tbl <- read.csv(file.path(report_dir, "registration_summary.csv"), check.names = FALSE)
history_path <- file.path(report_dir, "registration_history_events.csv")
history_events <- if (file.exists(history_path)) {
  read.csv(history_path, check.names = FALSE)
} else {
  data.frame()
}

fetch_metadata_path <- file.path(data_dir, "metadata", "registration_registry_fetch_metadata.csv")
fetch_metadata <- if (file.exists(fetch_metadata_path)) {
  read.csv(fetch_metadata_path, check.names = FALSE)
} else {
  data.frame()
}

current_registry_path <- file.path(data_dir, "inputs", "registration_registry_current.csv")
current_registry <- if (file.exists(current_registry_path)) {
  read.csv(current_registry_path, check.names = FALSE)
} else {
  data.frame()
}

compact_character <- function(x, width = 44) {
  x <- as.character(x)
  x[is.na(x)] <- ""
  too_long <- nchar(x) > width
  x[too_long] <- paste0(substr(x[too_long], 1, width - 3), "...")
  x
}

compact_df <- function(df, cols = NULL, n = 20, char_width = 44) {
  out <- df
  if (!is.null(cols)) {
    keep <- intersect(cols, names(out))
    out <- out[, keep, drop = FALSE]
  }
  if (nrow(out) > n) {

```

```

    out <- utils::head(out, n)
  }
  is_char_like <- vapply(out, function(col) is.character(col) || is.factor(col), logical(1))
  out[is_char_like] <- lapply(out[is_char_like], compact_character, width = char_width)
  out
}

```

3 Interpretation Key

- **n_claims**: total claim rows evaluated, including report-protocol and registry-derived rows.
- **n_claims_assessed**: rows with enough information to classify as **match** or **mismatch**.
- **n_mismatch**: assessed rows classified as mismatch.
- **mismatch_rate**: $\text{n_mismatch} / \text{n_claims_assessed}$; indeterminate and not-assessed rows are excluded.
- **n_missing**: rows with missing report or protocol values.
- **missing_rate**: missing report/protocol burden across all claim rows.
- **prospective_registration_status**: whether the registry first-posted date appears on or before the registered study start date.
- **n_major_registry_history_changes**: local history-event rows marked medium/high severity.
- **results_overdue_flag**: registry results appear absent more than 365 days after primary completion in this screen.
- **publication_link_missing_flag**: a supplied DOI/PMID/URL was not found in registry references.

Classification labels:

- **match**: available extracted values agree under the implemented screen.
- **mismatch**: available extracted values conflict or a prespecified registry screen fails.
- **indeterminate**: the source contains insufficient or ambiguous information for automated classification.
- **not_assessed**: the screen is not applicable or no eligible registry/source data were available.

```
knitr::kable(summary_tbl, digits = 4)
```

	n_in- de- ter-				prospec- tive_reg- istra- tion_sta-				n_ma- jor_reg- istry_his- tory_changes	re- sults_over- use_flag	publi- ca- tion_link_miss- ing_flag	
	n_claims	n_solutions	not mis-	n_mis-	n_miss-	n_miss-	n_miss-	n_miss-				
trial_id	claims	not mis-	n_mis-	n_miss-	n_miss-	n_miss-	n_miss-	n_miss-				
lung-18 time	18	16	1	1	2	0.125	1	0.0556	prospec- tive	0	TRUE	FALSE

4 Registry Source

```
if (nrow(fetch_metadata) == 0) {
  cat("No registry fetch metadata was available.\n")
} else {
  knitr::kable(
    compact_df(
      fetch_metadata,
      cols = c(
        "registry_id",
        "registry_id_source",
        "registry_current_source",
        "allow_network",
        "fetch_status",
        "fetch_message",
        "api_url"
      ),
      n = 5,
      char_width = 42
    )
  )
}
```

reg- istry_id	reg- istry_id	reg- istry_cur- rent_source	al- low_net-work	fetch_sta- tus	fetch_message	api_url
NCT0549037	source_text	clinicaltrials_api_v2	True	fetching	Fetch current registry record from Cl...	https://clinicaltrials.gov/api/v2/studi...

```
if (nrow(current_registry) > 0) {
  knitr::kable(
    compact_df(
      current_registry,
      cols = c(
        "registry_id",
        "study_type",
        "overall_status",
        "start_date",
        "study_first_posted_date",
        "primary_completion_date",
        "has_results",
        "enrollment_count",
        "allocation",
        "masking"
      ),
      n = 5,
      char_width = 36
    )
  )
} else {
  cat("No normalized ClinicalTrials.gov current record was available.\n")
}
```

reg- istry_id	study_type	over- all_sta- tus	start_date	study_first_ posted_date	pri- mary_com- ple- tion_date	has_re- sults	enroll- ment_count	allo- ca- tion	mask- ing
NCT055119037	INTER- VEN- TIONAL	RE- CRUIT- ING	2022- 09- 23	2022-09-22	2024-12-02	False	210	RAN- DOM- IZED	NONE

5 Report-Protocol Claims

```
report_protocol <- row_results[row_results$claim_category == "report_protocol", , drop = FALSE]
knitr::kable(
  compact_df(
```

```

report_protocol,
cols = c(
  "claim_id",
  "report_value",
  "protocol_value",
  "assessment_status",
  "page_ref",
  "severity"
),
n = 25,
char_width = 38
)
)

```

claim_id	re- port_value	proto- col_value	assess- ment_status	page_ref	sever- ity
reg- istry_id_over- lap	NCT05549037		mismatch	report_page=1; protocol_page=	medium
allocation_ra- tio	1:1	1:1	match	report_page=1; protocol_page=9.0	info
randomiza- tion_phrase	present	present	match	report_page=2; protocol_page=6.0	info
blind- ing_phrase	present	present	match	report_page=2; protocol_page=6.0	info

6 ClinicalTrials.gov Current-Record Claims

```

registry_claims <- row_results[row_results$claim_category != "report_protocol", , drop = FALSE]
if (nrow(registry_claims) == 0) {
  cat("No ClinicalTrials.gov-derived claim rows were available.\n")
} else {
  knitr::kable(
    compact_df(
      registry_claims,
      cols = c(
        "claim_id",
        "assessment_status",

```

```

        "severity",
        "report_value",
        "registry_value",
        "notes"
    ),
    n = 30,
    char_width = 34
)
}

```

	claim_id	as- sess- ment_status	severity	report_value	registry_value	notes
5	clinicaltrials_registry_id_cons...	match	info	NCT05549037	NCT05549037	Compares NCT IDs extracted from...
6	clinicaltrials_prospective_regi...	match	info		start=2022-09-23;	Prospective if the latest
7	clinicaltrials_enrollment_congr...	match	info	01 1 2 3 6 7 9 12 14 16 17 18 2...	first_posted=... 210 (ESTIMATED)	possi... Registry enrollment was found i...
8	clinicaltrials_primary_outcome_...	match	info		Progression-free survival (PFS)...	First registry primary outcome ...
9	clinicaltrials_secondary_outcome...	match	info		Objective Response Rate (ORR) ...	First registry secondary outcom...
10	clinicaltrials_allocation_congr...	match	info		RANDOMIZED	Registry design term found in s...
11	clinicaltrials_masking_congruence	match	info		NONE	Registry design term found in s...

	claim_id	as-session-status	ever-ity	report_value	registry_value	notes
12	clinicaltrials_intervention_mod...	match	info		PARALLEL	Registry design term found in s...
13	clinicaltrials_arms_congruence	match	info		Arm A: Morning group EXPERIME...	First registry label found in s...
14	clinicaltrials_interventions_co...	match	info		Pem-brolizumab, Sintilimab DRU...	First registry label found in s...
15	clinicaltrials_results_posted	indefinite	info		has_results=False; results_firs...	Presence/absence of posted regi...
16	clinicaltrials_results_overdue	mis-match	high		primary_completion=2024-12-02; ...	Uses a 365-day screen after pri...
17	clinicaltrials_publication_linkage	not_assessed	info		41629425 DERIVED Huang Z, Z...	Checks whether supplied publica...
18	clinicaltrials_current_record_a...	match	info		NCT05549037	Fetches current registry record...

7 Registry History Events

Registry history is optional in this version. If no local history file is supplied, this table is expected to be empty.


```

if (nrow(history_events) == 0) {
  cat("No local registry-history events were available.\n")
} else {
  knitr::kable(
    compact_df(
      history_events,
      cols = c(
        "event_date",
        "registry_field",
        "old_value",
        "new_value",
        "severity",
        "notes"
      ),
      n = 30,
      char_width = 34
    )
  )
}

```

No local registry-history events were available.